SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0714

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario-Based

Code of Conduct:

I Jasmitha R(192524295)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
JASMITHA R

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

Given Network: 192.168.100.0/24

The available IP block contains **256 IP addresses (192.168.100.0 – 192.168.100.255)**.

Department	Subnet Mask	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
School of Computing	/25 (255.255.255.128)	192.168.100.0	192.168.100.1 – 192.168.100.126	192.168.100.127	126
School of Electronics	/26 (255.255.255.192)	192.168.100.128	192.168.100.129 – 192.168.100.190	192.168.100.191	62
School of Mechanical Engineering	/27 (255.255.255.224)	192.168.100.192	192.168.100.193 – 192.168.100.222	192.168.100.223	30

1. Network Address and Broadcast Address

- **School of Computing**
    - Network Address: 192.168.100.0

    - Broadcast Address: 192.168.100.127
- **School of Electronics**
    - Network Address: 192.168.100.128
    - Broadcast Address: 192.168.100.191
- **School of Mechanical Engineering**
    - Network Address: 192.168.100.192
    - Broadcast Address: 192.168.100.223

2. Valid Host IP Address Range

- School of Computing: 192.168.100.1 – 192.168.100.126
- School of Electronics: 192.168.100.129 – 192.168.100.190
- School of Mechanical Engineering: 192.168.100.193 – 192.168.100.222

3. Number of Usable Host Addresses

- **/25:** (2^{7} - 2 = 126) usable hosts
- **/26:** (2^{6} - 2 = 62) usable hosts
- **/27:** (2^{5} - 2 = 30) usable hosts

4. Remaining Unused IP Address Range

The allocated subnets use addresses from 192.168.100.0 to 192.168.100.223.

Unused Range: 192.168.100.224 – 192.168.100.255

- Network Address of remaining block: 192.168.100.224/27
- Broadcast Address: 192.168.100.255
- Valid Host Range: 192.168.100.225 – 192.168.100.254
- Usable Hosts Available: 30

Conclusion

The 192.168.100.0/24 network is successfully divided into three subnets:

- **School of Computing:** 126 hosts
- **School of Electronics:** 62 hosts
- **School of Mechanical Engineering:** 30 hosts

The remaining subnet 192.168.100.224/27 is left unused and can be reserved for future expansion of the Smart Campus network.

2.A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

ANSWER:

Given Network: 10.10.0.0/16

The **10.10.0.0/16** network contains **65,536 IP addresses** (10.10.0.0 – 10.10.255.255).

Subnet Allocation Table

Department	Subnet Mask	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
Software Development	/24 (255.255.255.0)	**10.10.0.0**	**10.10.0.1 – 10.10.0.254**	**10.10.0.255**	**254**
Quality Assurance	/25 (255.255.255.128)	**10.10.1.0**	**10.10.1.1 – 10.10.1.126**	**10.10.1.127**	**126**
Human Resources	/26 (255.255.255.192)	**10.10.1.128**	**10.10.1.129 – 10.10.1.1**	**10.10.1.191**	**62**

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Executive Management	/27 (255.255.255.224)	**10.10.1.192**	**10.10.1.193 – 10.10.1.222**	**10.10.1.223**	**30**

1. Network Address and Broadcast Address

Software Development (/24)

- **Network Address:** 10.10.0.0
- **Broadcast Address:** 10.10.0.255

Quality Assurance (/25)

- **Network Address:** 10.10.1.0
- **Broadcast Address:** 10.10.1.127

Human Resources (/26)

- **Network Address:** 10.10.1.128
- **Broadcast Address:** 10.10.1.191

Executive Management (/27)

- **Network Address:** 10.10.1.192
- **Broadcast Address:** 10.10.1.223

2. Valid Host IP Address Range

- Software Development: 10.10.0.1 – 10.10.0.254
- Quality Assurance: 10.10.1.1 – 10.10.1.126
- Human Resources: 10.10.1.129 – 10.10.1.190
- Executive Management: 10.10.1.193 – 10.10.1.222

3. Number of Usable Hosts

- **/24:** (2^8 - 2 = 254) usable hosts
- **/25:** (2^7 - 2 = 126) usable hosts
- **/26:** (2^6 - 2 = 62) usable hosts

- **/27:** (2^5 - 2 = 30) usable hosts

4. Remaining Unused IP Address Range

The allocated subnets occupy addresses from:

10.10.0.0 → 10.10.1.223

Unused IP Address Range:

- **10.10.1.224 – 10.10.255.255**

This remaining range can be reserved for future departments or network expansion.

Conclusion

The **10.10.0.0/16** network is divided as follows:

- **Software Development:** 254 usable hosts
- **Quality Assurance:** 126 usable hosts
- **Human Resources:** 62 usable hosts
- **Executive Management:** 30 usable hosts

The remaining address range **10.10.1.224 – 10.10.255.255** is available for future subnet allocation. This subnet design ensures efficient IP address utilization and supports future expansion of the company's network.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

ANSWER:

Given IPv6 Network

2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. Convert the Given IPv6 Address into Compressed Notation

IPv6 allows consecutive **0000** blocks to be replaced with **::**.

Compressed IPv6 Address:

2001:db8:abcd::/64

2. Expand the Compressed Address into Full Hexadecimal Representation

The compressed address **2001:db8:abcd::/64** expands to:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

3. Identify the Network Prefix Using the /64 Prefix Length

- **Prefix Length: /64**
- **Network Prefix: 2001:0db8:abcd:0000::/64**

The first **64 bits** represent the **network**, and the last **64 bits** represent the **host/interface ID**.

4. Calculate the Total Number of Host Addresses Supported

A **/64** IPv6 subnet leaves **64 bits** for host addresses.

Formula:

Host Addresses = 2^{64}

= 18,446,744,073,709,551,616 host addresses

5. Total Number of Possible Host Addresses Available

Since IPv6 does not reserve broadcast addresses, **all 2^{64} addresses are available**.

Total Possible Host Addresses:

18,446,744,073,709,551,616 (2^{64})

Conclusion

The hospital's IPv6 network **2001:db8:abcd::/64** supports an extremely large number of devices, making it ideal for IoT sensors, medical equipment, and future network expansion. The **/64 prefix** is the standard subnet size in IPv6 and provides **2^{64} (18.4 quintillion)** possible host addresses.

4.An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The matching routing table entry.
3. The **next-hop network/interface** used to forward the packet.

ANSWER:

Introduction

A router is a Layer 3 (Network Layer) device that forwards packets from one network to another based on the destination IP address. It checks its routing table and selects the best matching route using the Longest Prefix Match (LPM) rule. In this scenario, the router receives a packet with destination IP 192.168.2.45 and determines where to forward it.

Step 1: Analyze the Destination IP Address

Destination IP Address: 192.168.2.45
Subnet Mask: 255.255.255.0 (/24)
A /24 subnet mask means:
First 24 bits represent the Network ID.
Last 8 bits represent the Host ID.
Binary Representation
Destination IP Address
192.168.2.45

11000000.10101000.00000010.00101101
Subnet Mask (/24)
255.255.255.0

11111111.11111111.11111111.00000000
Performing the AND operation:
IP Address:
11000000.10101000.00000010.00101101

Subnet Mask:
11111111.11111111.11111111.00000000

Network:
11000000.10101000.00000010.00000000
Network Address = 192.168.2.0
Therefore, the packet belongs to the 192.168.2.0/24 subnet.

Step 2: Compare with the Routing Table
The router compares the destination network with all routing table entries.

Routing Table Entry	Match?
192.168.1.0/24	No
192.168.2.0/24	Yes
192.168.3.0/24	No

The destination network exactly matches 192.168.2.0/24.

Step 3: Determine the Matching Routing Entry
The router selects:
Destination Network: 192.168.2.0/24
Because it is the exact match for the destination IP address.
This is also called the Longest Prefix Match, where the router always selects the route with the most specific subnet mask.

Step 4: Determine the Next-Hop Network / Interface
After finding the matching route, the router forwards the packet through the interface connected to the 192.168.2.0/24 network.
Example:

Destination	Next Hop	Outgoing Interface
192.168.2.0/24	Directly Connected	GigabitEthernet0/1

If the network is directly connected, the router sends the packet directly through GigabitEthernet0/1.
If it is a remote network, the router forwards it to the configured next-hop router.

Step 5: Packet Forwarding Process
The packet forwarding process is:
Router receives the packet.
Reads the destination IP address (192.168.2.45).
Applies the subnet mask (/24).
Finds the network address (192.168.2.0).
Searches the routing table.
Finds the matching route (192.168.2.0/24).
Forwards the packet through the interface connected to 192.168.2.0/24.

Advantages of Routing

- Delivers packets to the correct destination.
- Reduces unnecessary network traffic.
- Supports communication between multiple networks.
- Uses routing tables for efficient packet forwarding.
- Ensures reliable and scalable network communication.

Conclusion

The router receives a packet with destination IP 192.168.2.45 and performs a routing table lookup. After applying the /24 subnet mask, it identifies the destination network as 192.168.2.0/24.

4. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

Answer

Introduction

A **Layer-3 Router** connects different Local Area Networks (LANs) and enables communication between them. Each LAN has its own **network address, broadcast address, host range, and default gateway**. The default gateway is the router interface through which devices communicate with other networks.

Step 1: Determine the Network Address and Broadcast Address

Administration LAN

Subnet: 192.168.10.0/26

Subnet Mask: 255.255.255.192

Block Size: 64 addresses

Network Address: 192.168.10.0

Broadcast Address: 192.168.10.63

Finance LAN

Subnet: 192.168.10.64/26

Subnet Mask: 255.255.255.192

Network Address: 192.168.10.64

Broadcast Address: 192.168.10.127

Step 2: Identify the Valid Host IP Address Range

Administration LAN

First Host: **192.168.10.1**

Last Host: **192.168.10.62**

Valid Host Range:

192.168.10.1 – 192.168.10.62

Finance LAN
First Host: **192.168.10.65**
Last Host: **192.168.10.126**
Valid Host Range:
192.168.10.65 – 192.168.10.126

Step 3: Suggest Suitable IP Addresses for Three Computers
Administration LAN

Computer	IP Address
Computer 1	**192.168.10.2**
Computer 2	**192.168.10.3**
Computer 3	**192.168.10.4**

(The router uses **192.168.10.1** as the gateway.)

Finance LAN

Computer	IP Address
Computer 1	**192.168.10.66**
Computer 2	**192.168.10.67**
Computer 3	**192.168.10.68**

(The router uses **192.168.10.65** as the gateway.)

Step 4: Identify the Default Gateway
The **default gateway** is the IP address of the router interface connected to each LAN.

LAN	Default Gateway
Administration LAN	**192.168.10.1**
Finance LAN	**192.168.10.65**

All hosts in each LAN must use these gateway addresses to communicate with other networks.

Advantages of Using a Router
Connects multiple LANs and enables inter-network communication.
Separates broadcast domains, improving network performance.
Provides secure communication between departments.
Supports efficient IP address management and routing.
Allows future expansion by adding more subnets.

Conclusion
The company network is divided into two **/26 subnets**: **Administration LAN** and **Finance LAN**, each supporting **62 usable host addresses**.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand